

DAVIDE MIRZA

Medford, Massachusetts

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[LinkedIn](#) [Portfolio](#)

Education

Tufts University – Medford, MA

Bachelor of Science, Mechanical Engineering | Minor, Engineering Management

Expected May 2027

- GPA 3.8/4.0 — Dean's List, all semesters
- Relevant Coursework: Mechanics, Thermal Fluid Systems, Electronics and Controls, Engineering Design, Materials and Manufacturing, General Physics with Lab, Technical and Managerial Communication

Professional Experience

SharkNinja

June 2026 – August 2026

Mechanical Engineering Intern – New Product Development (NPD) – Needham, MA

- Owned structural architecture work on a new frozen-treat product: redesigned the housing load path to cut the moment about the gearbox center, and correlated vertical motor current to frame load by stalling the DC drive into a 10 kN load cell, so any recipe run's structural load could be inferred from current alone
- Built a computer-vision method using ArUco markers to measure C-frame opening from video — contactless, sub-millimeter, self-calibrating — then reran it on every structural hack and CAD revision, turning stiffness from a subjective judgment into a repeatable measured comparison
- Validated the CAD revisions on hardware by running 10 recipes across springs spanning 10–30 N, selecting the preload that kept the blade from stalling on the hardest recipes while still meeting KPIs on freeze-point detection, cycle time, and unit-to-unit repeatability
- Partnered with Product Design on user touchpoints — cup retention, insertion feel, and rattle — translating ID intent into a bayonet-retained bushing assembly modeled in SolidWorks with tolerancing and printed fit validation

Loccioni

June 2025 – August 2025

Robotics / Mechanical Engineering Intern – APOTECABag – Angeli di Rosora, Italy

- Designed in Solid Edge a more advanced APOTECABag robot meeting stricter GMP practices, with a new floor-to-ceiling (FTC) implementation for operation in the Mayo Clinic cleanroom
- Conducted a feasibility study of the APOTECABag robot to identify the minimum distance between the air measuring probe and the dosage area of the machine, a GMP requirement
- Assembled and tested 8 functional units based on 2D and 3D technical drawings and integrated them into the APOTECABag main frame; identified 10–12 potential issues and proposed improvements
- Designed a custom testing bench for verifying the sealing of the robot's cooling circuit, interpreting pneumatic schematics

Engineering Projects

Waffle House — Automated Waffle Production Line

2026

- Designed, built and programmed the whipped cream and maple syrup dispensing station of a fully automated waffle line: a stepper-actuated can dispenser driven by a DC rack and pinion, and a syringe dispenser with a stepper-driven plunger
- Added an anti-drip mechanism that reverse-rotates the syringe stepper after each cycle to retract the plunger and prevent dripping onto the plate
- Ran the station on two Raspberry Pis polling an Airtable database over API, dispensing only once the mobile iRobot Create 3 platform arrived carrying the matching order

Create 3 Object Recognition — Autonomous Pick-and-Place Robot

2026

- Built an autonomous warehouse-style robot on an iRobot Create 3 base that locates, identifies, picks up and delivers figures with no human intervention, under unknown object orientation and drop-off locations revealed only shortly before the demo
- Combined ROS 2 (rclpy) waypoint navigation with a trained neural-network classifier running on an onboard Raspberry Pi camera to identify each figure and select its matching drop-off location
- Designed and integrated a stepper-motor-driven gripper for pick-up and release

Line Follower — Custom-Fabricated Robot

2026

- Built a line-following robot on a fully hand-fabricated chassis of laser-cut and hand-cut wooden plates, drilled and spaced for motor and electronics mounting, with 3D-printed brackets locating the IR sensor array
- Programmed a Raspberry Pi to read a three-sensor IR array and drive the motors through an L298N driver using PWM speed and direction control
- Implemented an end-of-path turn-around routine that re-acquires and re-centers on the line to complete the return leg autonomously

Skills

CAD & Drawings: SolidWorks, Solid Edge, AutoCAD, Onshape; parametric and surface modeling, sheet metal, 20+ component assemblies; press, clearance and threaded fits; surface finish (Ra), thread and weld callouts; snap fits, hinges, sliders, detents, bayonet retention

Machining & DFM: Manual mill and lathe, reaming, thread cutting; CNC and CAM programming, G-code; custom fixture, test-bench and workholding design; DFM and feasibility studies for production parts; aluminum anodizing, steel finishing

Fabrication & Materials: FDM and SLA printing, laser cutting, woodworking, mechanical assembly and soldering; injection molding, vacuum forming, casting, TIG welding; aluminum, steel, Delrin, ABS, wood

Metrology & Test: Test rig and fixture design, test protocol design and repeatability studies; load cells, motor current sensing, data logging; computer-vision metrology (ArUco, OpenCV); calipers, micrometers, gauges

Electromechanical: Arduino (C++), Raspberry Pi, soldering; DC and stepper motors, encoders, PWM and L298N motor drivers; IR, color, camera and load-cell sensing; PCB design (KiCad) and integration

Software: Python, C++, MATLAB, ROS 2 (rclpy), OpenCV, Keras, COMSOL, Airtable API, Excel, Figma, JD Edwards

Languages: Fluent in Italian, Romanian and English

Leadership & Athletics

Tufts University Varsity Tennis – Captain, Singles and Doubles

2023 – Present

- Captain of a 16-player varsity roster; train 20–25 hours a week including on-court practice, fitness and conditioning
- Compete in tournaments and matches against DIII colleges throughout the fall and spring seasons